

Exp no :04

Date :

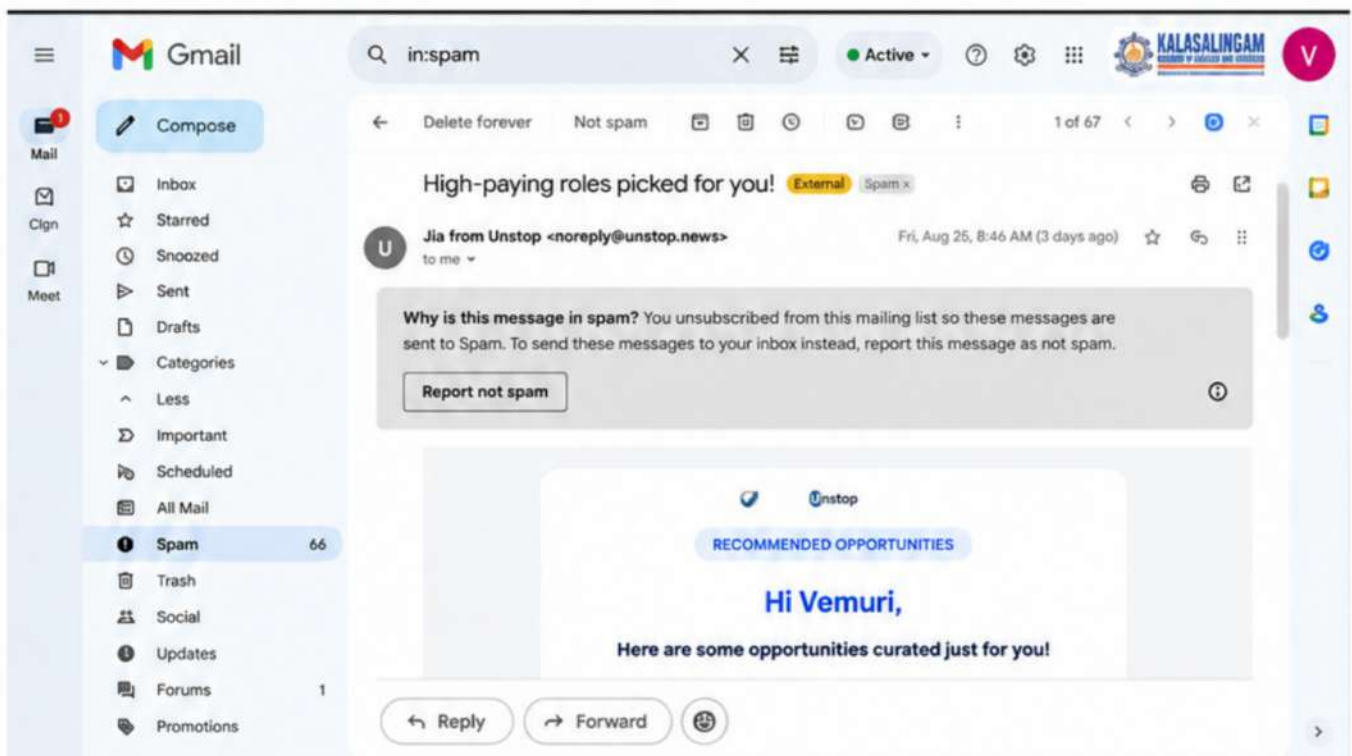
Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.



Whois IP 103.52.180.165

Updated 6 days ago

```
% [whois.apnic.net]
% whois data copyright terms    http://www.apnic.net/db/dbcopyright.html

% Information related to '103.52.180.0 - 103.52.183.255'

% Abuse contacts for '103.52.180.0 - 103.52.183.255' are 'abuse@mumbai.ravience.in', 'abuse@netcorecloud.com'

inetnum:        103.52.180.0 - 103.52.183.255
netname:        NETCORE-IN
descr:          Netcore Solutions Pvt Ltd
descr:          8th Floor, Peninsula Tower, Peninsula Corporate Park, G.K. Marg, Lower Parel (W), Mba
admin-c:        PWA78-AP
tech-c:         KJ35-AP
country:        IN
mnt-by:         MAINT-IN-IRINN
mnt-lower:      MAINT-IN-NETCORE-INC
mnt-lower:      MAINT-IN-IRINN
mnt-routes:     MAINT-IN-NETCORE-INC
mnt-routes:     MAINT-IN-IRINN
mnt-irt:        IRT-NETCORE-IN
status:         ALLOCATED PORTABLE
last-modified:  2025-08-11T22:49:15Z
source:         APNIC

irt:            IRT-NETCORE-IN
address:        402, PENINSULA CHAMBERS,
address:        G.K. MARG,
address:        LOWER PAREL (W),
address:        MUMBAI - 400013
e-mail:         admin@mumbai.ravience.in
e-mail:         kuldeep@netcorecloud.com
abuse-mailbox:  abuse@mumbai.ravience.in
abuse-mailbox:  abuse@netcorecloud.com
```

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.
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Google Admin Toolbox Messageheader

Delivered-To: 99240040333@klu.ac.in
Received: by 2002:adf:eac8:0:b0:482:c3f2:4d63 with SMTP id o8csp392792wm; Fri, 09 Jan 2026 23:30:00 -0700 (PST)
X-Received: by 2002:a05:622a:30c:b0:52f:a2c0:1451 with SMTP id d75a77b69052e-52fb93d17e4mr45253971d7.71787888960771; Fri, 09 Jan 2026 23:30:00 -0700 (PST);
ARC-Seal: i=1; a=rsa-sha256; t=1787868960; cv=none; d=google.com; s=arc-20260327; b=jl4crIDeGhdqIFnR/XQUpzPHBWPbtrppPPCmVniBjBC+U7jhpvgUYPZ9z1hz87jKZ9u72xRgSleZQyQmml0nqe+Y3JRQqXYU0m+PISYB7h61iVfdPLOxKiBrsbGHAYX3n+96bYnjtL0b57SAulc7yghbX+9rYEo419dUBY/RG+a1i0iTufnQ90a67g8L8Xqvi2yo8meilMQkWyMNAqPJSr9WTf7FIQY3AorJ0FWpbQ4VvMhKKDVSioKUJXDSkYVeLu4NCMD0e1+qe+U5JrDQ3SKsuGoV1Fdv/hk1J3hoCANaBtm/cpHkOhxwzbMu5SOoZHYmjdHP 4uLgp==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=date:message-id:list-unsubscribe:post-list-unsubscribe:feedback-id:reply-to:from:subject:mime-version:dkim-signature:dkim-signature; bh=0/29kTwQhXEeSw6PhcSPAk1Wa2mVTxRJB7UWyhbXlnw=;

[ANALYZE THE HEADER ABOVE](#)



Google Admin Toolbox Messageheader

[Help](#)

Messageid	20260826013939.63EE0EB869@campaign-231.appspot.com
Created at:	09/01/2026, 11:30:00 PM GMT+5:30 (Delivered after 1 sec)
From:	Jia from Unstop <noreply@unstop.news> (using SendGrid (now Twilio))
To:	99240040333@klu.ac.in
Subject:	High-paying roles picked for you!
DKIM:	pass with domain unstop.news Learn more

#	Delay	From *	To *	Protocol	Time received
1		100.100.smtpout.na.tl.sendgrid.net	→ Bmaprf: mx.google.com	ESMTPS	09/01/2026, 11:29:59 PM GMT+5:30
2		2002:a05:622a:30c:b0:52f:a2c0:1451	→ Bmaprf: 2002:adf:eac8:0:b0:482:c3f2:4d63	SMTP	09/01/2026, 11:30:00 PM GMT+5:30
3	1 sec		→ Bmaprf: 2002:adf:eac8:0:b0:482:c3f2:4d63	SMTP	09/01/2026, 11:30:00 PM GMT+5:30

[ANALYZE ANOTHER HEADER](#)

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.